

Part A: Order of Operations*Evaluate each expression completely.*

1. $(-9 + (-2))^2 \div (-11)$

9. $(-2 + 15)^2 \div 1$

2. $(5 + 3)^2 \div (-2)$

10. $24 - 4 \cdot ((-9) + 24)$

3. $(-5)^2 - (-20) \cdot 6$

11. $31 - 8 \cdot ((-5) + 31)$

4. $(-4 + 12)^2 \div 2$

12. $14 - 3 \cdot ((-2) + 14)$

5. $(9 + 3)^2 \div (-8)$

13. $8^2 - 3 \cdot 4$

6. $(2 + 14)^2 \div (-8)$

14. $40 - 4 \cdot (5 + 40)$

7. $18 - 8 \cdot ((-4) + 18)$

15. $(-7)^3 - 17 \cdot 9$

8. $8^2 - 4 \cdot 2$

Part B: Negative Number Operations*Evaluate each expression completely.*

1. $-12 \cdot (-4)$

4. $-64 \div 8$

2. $-9 \cdot 11$

5. $-20 \div 10$

3. $8 \div (-4)$

6. $-60 \div 12$

7. $-34 + (-9) \cdot 15$

12. $-27 + 9 \cdot (-10)$

8. $37 + 8 \cdot 6$

13. $25 \div 5 - (-5)$

9. $11 + (-11) \cdot 6$

14. $72 \div (-6) - (-20)$

10. $196 \div 14 - (-32)$

15. $33 \div 3 - 40$

11. $28 + (-15) \cdot (-14)$

Part C: Solving Linear Equations

Solve each equation for x .

1. $9x - 10 = -\frac{13}{4}$

7. $5(4x + 9) + 4 = -191$

2. $12x + 12 = -54$

8. $8(x - 4) + 5 = 5$

3. $4x + 4 = -4$

9. $3(2x + 3) - 10 = -25$

4. $9x - 12 = 6$

10. $8(7x - 7) - 1 = 167$

5. $5x - 8 = -3$

11. $-6x + 14 = 11x - \frac{23}{2}$

6. $8(8x - 11) + 3 = -213$

12. $-8x + 13 = 15x + 151$

13. $-10x - 10 = -15x + 45$

15. $3x + 7 = 2x + 17$

14. $7x + 4 = -3x + 44$

Part D: Linear Inequalities

Solve each inequality for x . Remember to reverse the inequality sign when you multiply or divide by a negative number.

1. $x - 6 > -7$

9. $-12x + 7 < -125$

2. $5x \leq 35$

10. $6x - 2 \leq 7x - 1$

3. $x + 9 < 1$

11. $-9x + 13 < 4x + 65$

4. $2x - 3 < -21$

12. $-13x - 3 > -15x + 7$

5. $10x - 10 > -100$

13. $4x - 1 \leq \frac{1}{2}x - \frac{9}{2}$

6. $8x - 8 \leq 80$

14. $-2x - 12 < -\frac{5}{3}x - 10$

7. $-2x + 1 \geq 21$

15. $\frac{5}{3}x + 10 < 3x + \frac{82}{3}$

8. $2x + 9 > -9$

Part E: Slope-Intercept Form

Rewrite each equation in slope-intercept form if needed, then identify the slope m and the y -intercept b .

1. $y = 8x$

8. $-27x + 3y = -12$

2. $y = \frac{9}{2}x + 4$

9. $6y = 15x - 6$

3. $y = 3x - 5$

10. $4y = 36x - 16$

4. $y = 7x + 1$

11. $4y = 8x - 16$

5. $y = 4x - 9$

12. $36x - 4y = -28$

6. $y = 7x - 2$

13. $3y = 18x - 6$

7. $y = 9x - 5$

14. $-9x + 3y = 24$

15. $4y = 14x - 24$

Part F: Exponent Rules

Simplify each expression. Write your answer using only positive exponents.

1. $4x^8 \cdot 9x^9$

3. $9x^3 \cdot 2x^2$

2. $3x^2 \cdot 6x^6$

4. $\frac{5x^6}{x^2}$

10. $7 \cdot (3^2)^{-2}$

5. $\frac{6x^6}{x^2}$

11. $\frac{5}{3^{-1}}$

6. $\frac{2x^7}{x^3}$

12. $9 \cdot 7^{-2}$

7. $(2x^7)^2$

13. $(2x^{-3})^3 \cdot x^{16}$

8. $(4x^2)^4$

14. $(2x^3y^3)^2 \cdot 2x^2y^8$

9. $(6x^2)^2$

15. $(6x^8y^{10})^3 \cdot 6x^{12}y^8$

Part G: Square and Cube Roots

Simplify each root completely. If the radicand is not a perfect square or cube, use prime factorization to pull out every factor you can.

1. $\sqrt{49}$

6. $\sqrt{150}$

11. $\sqrt{8}$

2. $\sqrt{64}$

7. $\sqrt{60}$

12. $\sqrt{40}$

3. $\sqrt{9}$

8. $\sqrt{124}$

13. $\sqrt[3]{250}$

4. $\sqrt[3]{27}$

9. $\sqrt{92}$

14. $\sqrt[3]{232}$

5. $\sqrt[3]{8}$

10. $\sqrt{148}$

15. $\sqrt[3]{243}$

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15. $\sqrt[3]{243}$

Answer Key

Part A: Order of Operations

- | | | |
|----------|-----------|------------|
| 1. -11 | 6. -32 | 11. -177 |
| 2. -32 | 7. -94 | 12. -22 |
| 3. 145 | 8. 56 | 13. 52 |
| 4. 32 | 9. 169 | 14. -140 |
| 5. -18 | 10. -36 | 15. -496 |

Part B: Negative Number Operations

- | | | |
|----------|-----------|------------|
| 1. 48 | 6. -5 | 11. 238 |
| 2. -99 | 7. -169 | 12. -117 |
| 3. -2 | 8. 85 | 13. 10 |
| 4. -8 | 9. -55 | 14. 8 |
| 5. -2 | 10. 46 | 15. -29 |

Part C: Solving Linear Equations

- | | | |
|------------------------|--------------|-----------------------|
| 1. $x = \frac{3}{4}$ | 6. $x = -2$ | 11. $x = \frac{3}{2}$ |
| 2. $x = -\frac{11}{2}$ | 7. $x = -12$ | 12. $x = -6$ |
| 3. $x = -2$ | 8. $x = 4$ | 13. $x = 11$ |
| 4. $x = 2$ | 9. $x = -4$ | 14. $x = 4$ |
| 5. $x = 1$ | 10. $x = 4$ | 15. $x = 10$ |

Part D: Linear Inequalities

- | | | |
|---------------|-----------------|-----------------|
| 1. $x > -1$ | 6. $x \leq 11$ | 11. $x > -4$ |
| 2. $x \leq 7$ | 7. $x \leq -10$ | 12. $x > 5$ |
| 3. $x < -8$ | 8. $x > -9$ | 13. $x \leq -1$ |
| 4. $x < -9$ | 9. $x > 11$ | 14. $x > -6$ |
| 5. $x > -9$ | 10. $x \geq -1$ | 15. $x > -13$ |

Part E: Slope-Intercept Form

- | | | |
|-----------------------------------|------------------------------------|-------------------------------------|
| 1. $m = 8, \quad b = 0$ | 6. $m = 7, \quad b = -2$ | 11. $m = 2, \quad b = -4$ |
| 2. $m = \frac{9}{2}, \quad b = 4$ | 7. $m = 9, \quad b = -5$ | 12. $m = 9, \quad b = 7$ |
| 3. $m = 3, \quad b = -5$ | 8. $m = 9, \quad b = -4$ | 13. $m = 6, \quad b = -2$ |
| 4. $m = 7, \quad b = 1$ | 9. $m = \frac{5}{2}, \quad b = -1$ | 14. $m = 3, \quad b = 8$ |
| 5. $m = 4, \quad b = -9$ | 10. $m = 9, \quad b = -4$ | 15. $m = \frac{7}{2}, \quad b = -6$ |

Part F: Exponent Rules

- | | | |
|---------------|--------------------|------------------------|
| 1. $36x^{17}$ | 6. $2x^4$ | 11. 15 |
| 2. $18x^8$ | 7. $4x^{14}$ | 12. $\frac{9}{49}$ |
| 3. $18x^5$ | 8. $256x^8$ | 13. $8x^7$ |
| 4. $5x^4$ | 9. $36x^4$ | 14. $8x^8y^{14}$ |
| 5. $6x^4$ | 10. $\frac{7}{81}$ | 15. $1296x^{36}y^{38}$ |

Part G: Square and Cube Roots

1. 7

2. 8

3. 3

4. 3

5. 2

6. $5\sqrt{6}$

7. $2\sqrt{15}$

8. $2\sqrt{31}$

9. $2\sqrt{23}$

10. $2\sqrt{37}$

11. $2\sqrt{2}$

12. $2\sqrt{10}$

13. $5\sqrt[3]{2}$

14. $2\sqrt[3]{29}$

15. $3\sqrt[3]{9}$